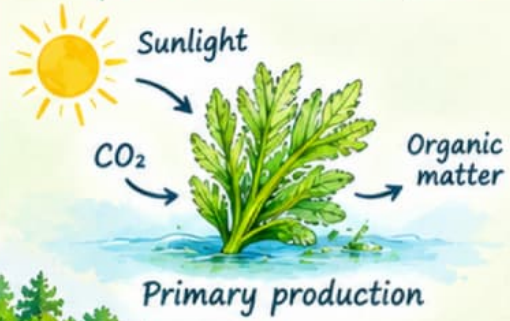
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Production and Productivity

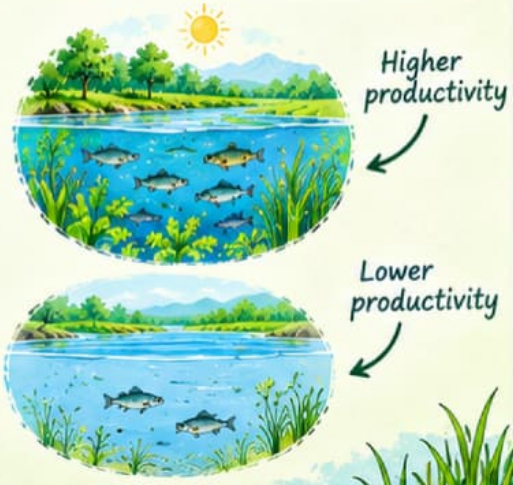
Production

The process of creating organic matter by producers (autotrophs).



Productivity

The rate at which organic matter is produced per unit time or area.



B.Sc. 2nd Semester | CBPBU

Created by
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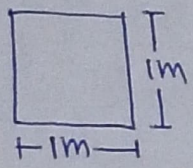
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1)

major function of an ecosystem/ Functional aspects of an ecosystem

- productivity
- Decomposition
- energy flow
- Nutrient cycling

production



→ Amount of biomass formation in $1m^2$

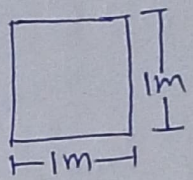
unit

Biomass / m^2

→ kg / m^2

→ Kcal / m^2

Productivity



→ Amount of biomass formatio in $1m^2$ in1year (fixed time

unit

Biomass / m^2. yr

→ kg m^{-2} yr^{-1}

→ Kcal m^{-2} yr^{-1}

✱ production in fixed time is c/o productivity.

production

primary production

→ Amount of Biomass formation by all producers of ecosystem

secondary production

→ Amount of biomass formation by all consumers of ecosystem.

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Productivity

Amount of biomass formation in $1m^2$ area in 1 year is called productivity.

or

The amount of organic matter accumulated in any unit time is c/o productivity.

Unit of productivity

→ Biomass m^{-2} yr^{-1}

→ Kg m^{-2} yr^{-1}

→ Kcal m^{-2} yr^{-1}

Types of productivity

primary productivity

→ The amount of biomass production by producers in $1 m^2$ in 1 year is c/o primary productivity

→ here solar energy converted into chemical energy or bond energy.

→ here producers ~~==~~ means autotrophs (green plants/ algae etc) produce the biomass

→ It is totally independent process.

Secondary productivity

→ The amount of biomass production by consumers in $1m^2$ in 1 year is c/o secondary productivity

→ here energy obtained from food consumed from Autotrophs

→ here consumers means heterotroph like herbivore, carnivore etc produce the biomass

→ This process depends on primary productivity

primary productivity

Gross primary productivity (GPP)

→ The total amount of biomass production by producers by photosynthesis including organic matter used in respiration

Net primary productivity (NPP)

→ The total amount of biomass production by ~~consume~~ producers after respiratory loss.

$$GPP - R_L = NPP$$

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∅ production v/s productivity

Basis of Difference	production	productivity
① Definition	The amount of biomass production in $1m^2$ area is c/o production	The amount of biomass production in $1m^2$ area in 1 year is c/o productivity.
② Meaning	Quantity produced	Efficiency of speed of production.
③ Time factor	Doesnot consider	consider
④ Nature	Absolute value	relative value
⑤ Measurement	measured by total biomass in unit area	measured by total biomass in unit area in unit time
⑥ Unit	$Kg\ m^{-2}$	$Kg\ m^{-2}\ yr^{-1}$
⑦ Ecological importance	Shows total output	shows efficiency of ecosystem.
⑧ example	A forest produce 5000 kg biomass	A forest produce 5000 kg biomass in 1 year

∅ primary productivity v/s secondary productivity

Basis of Difference	primary productivity	Secondary productivity
① Definition	Rate of production of organic matter by producer.	Rate of production of organic matter by consumer.
② organisms involved	producers	consumers
③ Source of energy	Solar energy	energy obtained from food
④ process	photosynthesis	consumption & assimilation of food.
⑤ trophic level	occurs at producer level	occurs at consumer level.
⑥ Dependency	Independent process	depend on primary productivity
⑦ Formation of food	producer can prepare food	Consumer cannot prepare food.

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⑧ importance	Form the base of food chain	Transfer energy to higher trophic level.
⑨ [illegible]		

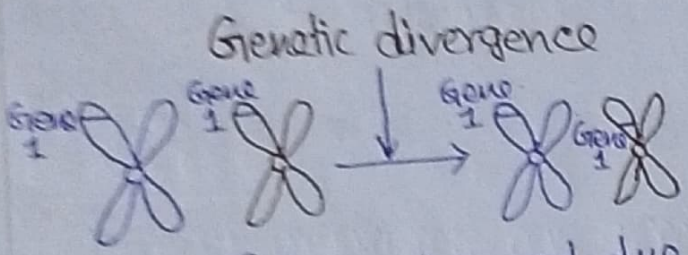
∅ GPP V/S NPP

Basis of Difference	GPP	NPP
① Definition	The total amount of biomass produced by producers including respiratory loss.	The total amount of biomass produced by producers after/excluding respirator loss.
② Meaning	Total energy fixed by plants	Actual stored energy availabe to consumer.
③ Respiration loss	Not excluded	excluded.
④ Energy content	Higher than NPP	lower than GPP
⑤ Availability to consumer	Not fully available	Available to herbivore & other carnivore.
⑥ Indicates	~~Totat~~ Total photosynthetic activity	Actual biomass accumulation.
⑦ Example	Plant produce 600 Kcal energy	After using 200 kcal 400 kcal energy availabe.

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Speciation–

Speciation is a evolutionary process by which new species are formed from an existing species through genetic divergence & reproductive isolation. It involves splitting of single ~~gene~~ species into two or three genetically isolated species.



location of Gene changed due to – mutation
– Genetic drift
– reduced gene flow
– Natural selection.
→ no crossing over → no meiosis

Reproductive isolation

prezygotic → before fertilization

post zygotic → after fertilization

⊛ Two different species cannot interbreed → ecological species concept

Types of Speciation

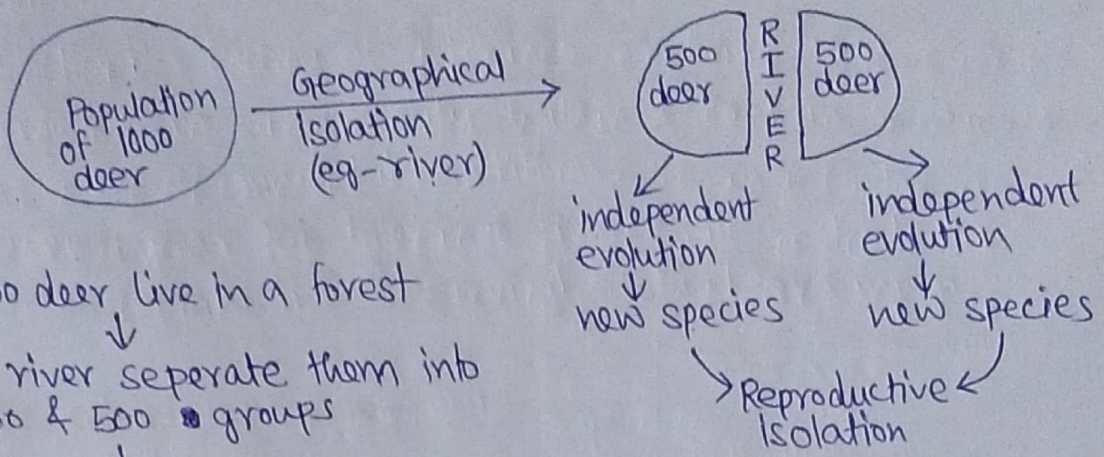
- Allopatric speciation
- Sympatric speciation
- parapatric speciation
- peripatric speciation

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① Allopatric speciation – (Geographical ~~distri~~ isolation)

When a population is seperated by Geographical barrier such as mountain, river, ocean; they isolated population evolve independently due to mutation, natural selection and become seperate species. (new species).

ex-



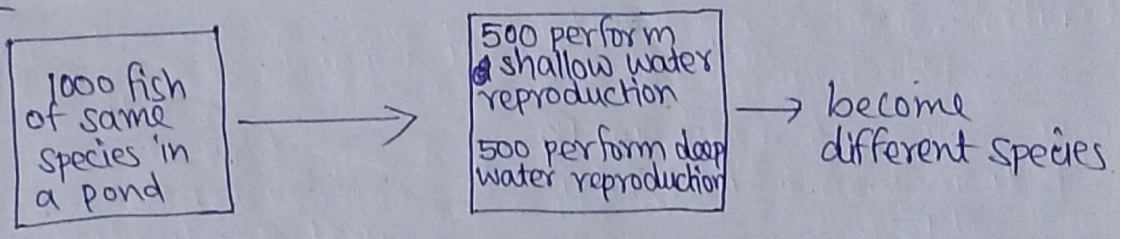
1000 deer live in a forest
↓
A river seperate them into 500 & 500 groups
↓
They perform independent evolution
↓
become two new species

② Sympatric speciation –

When species arise within the same geographic area due to genetic changes, is called sympatric species.

> There is no geographical barrier.

ex-



in a pond there was 1000 fish

500 perform shallow water reproduction
↓
new species

500 perform deep water reproduction
↓
new species

Reproductive isolation over time

para → ase pase
peri → international → dur dur

③ Parapatric speciation-

When ~~two~~ one species undergo any environmental stress but few of them ~~ev~~ genetically evolved & few remain as same, eventually they become different species but residing in same geographical area.

example-

100 butterfly in a field → Fungiside spray in one side of the field → 50 butterfly adopt resistance of it
→ 50 remain unchanged

→ They have limited gene ~~flo~~ flow
→ They remain in same geographic area.

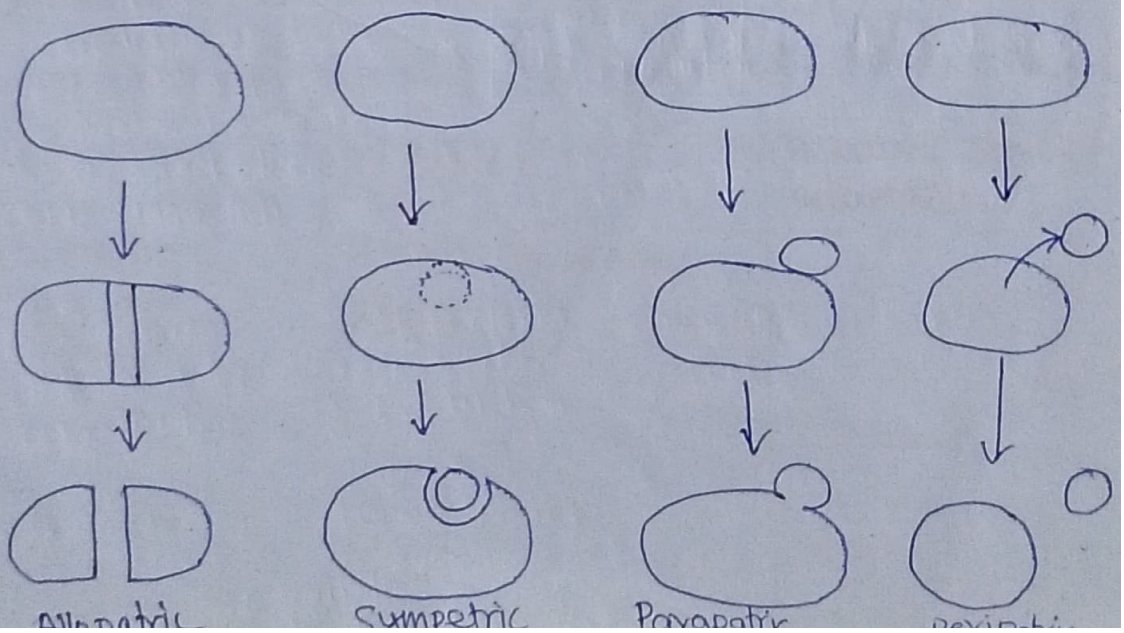
④ Peripatric speciation-

When some member of a ~~pop~~ population migrate to new habitat and the ~~cours~~ course of evolution they become different species is called peripatric speciation.

example

500 bird of a forest → 50 bird migrate to ~~isla~~ small island → they become different species during the caure of evolution.

Conclusion



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